

Beat-It: Beat-Synchronized Multi-Condition 3D Dance Generation

Zikai Huang¹, Xuemiao Xu^{1,3,4}, Cheng Xu², Huaidong Zhang^{1,3},
Chenxi Zheng¹, Jing Qin², and Shengfeng He⁵

¹ South China University of Technology, China
xuemx@scut.edu.cn

² The Hong Kong Polytechnic University, Hong Kong SAR, China
cschengxu@gmail.com

³ Guangdong Engineering Center for Large Model and GenAI Technology

⁴ Guangdong Provincial Key Lab of Computational Intelligence and Cyberspace
Information

⁵ Singapore Management University, Singapore
<https://zikaihuangscut.github.io/Beat-It/>

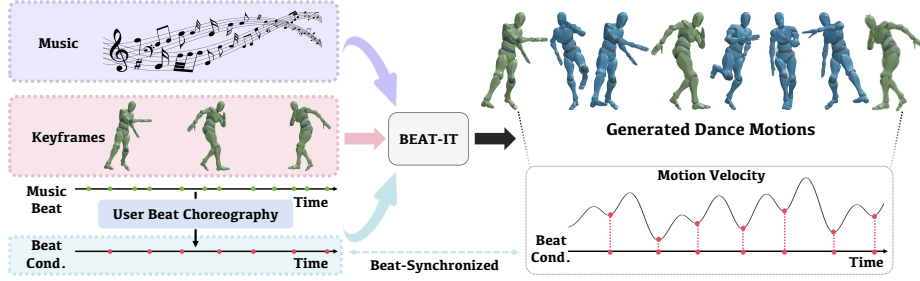


Fig. 1: We introduce *Beat-It*, a novel method for generating 3D dance motions with beat alignment and motion controllability. Our approach explicitly injects beat awareness and seamlessly integrates multiple conditions to guide the generation process, leading to beat-synchronized, key pose-guided dance generation.

Abstract. Dance, as an art form, fundamentally hinges on the precise synchronization with musical beats. However, achieving aesthetically pleasing dance sequences from music is challenging, with existing methods often falling short in controllability and beat alignment. To address these shortcomings, this paper introduces Beat-It, a novel framework for beat-specific, key pose-guided dance generation. Unlike prior approaches, Beat-It uniquely integrates explicit beat awareness and key pose guidance, effectively resolving two main issues: the misalignment of generated dance motions with musical beats, and the inability to map key poses to specific beats, critical for practical choreography. Our approach disentangles beat conditions from music using a nearest beat distance representation and employs a hierarchical multi-condition fusion mechanism. This mechanism

seamlessly integrates key poses, beats, and music features, mitigating condition conflicts and offering rich, multi-conditioned guidance for dance generation. Additionally, a specially designed beat alignment loss ensures the generated dance movements remain in sync with the designated beats. Extensive experiments confirm Beat-It’s superiority over existing state-of-the-art methods in terms of beat alignment and motion controllability.

Keywords: Dance generation · Beat synchronization · Multi-condition diffusion generation

1 Introduction

Dance, a time-honored art form, resonates across various ages and cultures due to its artistic and aesthetic significance, playing a vital role in the cultural and entertainment sectors. Traditional dance choreography creation, which demands an intricate balance of aesthetic movements, emotional expression, and precise synchronization with musical beats, often presents itself as a costly, laborious, and time-intensive endeavor, even for seasoned artists.

Recent advances in deep learning, particularly in generative models, have catalyzed efforts to automate the choreography process. Initial approaches predominantly focused on single-condition dance generation, learning a direct mapping from music to dance motions [1, 15, 21–23, 35, 39, 43]. The advent of diffusion models, renowned for their exceptional content generation capabilities, has marked a paradigm shift in this domain. For instance, Tseng et al. [42] introduced a transformer-based diffusion model that sets a new benchmark in music-to-dance conversion, showcasing the potential of these models in elevating the quality of generated dance sequences.

Notwithstanding the remarkable progress, achieving precise synchronization between dance motions and musical beats, coupled with flexible motion controllability in dance generation, remains a formidable challenge. In real-world choreography, the goal is often to assign specific key poses to particular musical beats and then fill in the connecting movements to form the complete dance sequence. This process requires not only the creation of designated key poses but also their harmonious synchronization with the musical beats. While few methods have explored key pose conditions [42, 52], none of them has considered explicit beat synchronization and controllability, rendering inferior dance motion generation. Furthermore, previous methods typically merge keyframe features with music features through simple concatenation or temporal/spatial blending. This approach, however, is flawed due to the sparse nature of keyframes. Consequently, these methods disproportionately emphasize dense music features at the expense of key pose conditions, leading to a notable misalignment between the generated dance motions and the specified keyframes.

To combat the above issues, we aim to completely disentangle beats from input music and simultaneously incorporate explicit beat and key pose guidance to enable beat-controllable, key pose-guided dance generation. We introduce Beat-It, a diffusion-based, multi-condition framework designed for this purpose (see

Fig. 1). Our approach begins by formulating beat condition as a nearest beat distance representation, which is a straightforward yet effective method to serve as beat guidance within our framework. To effectively harness and balance the multiple conditions involved, we propose a hierarchical multi-condition fusion mechanism. Initially, we integrate sparse key pose condition with dense beat and music conditions using a tailored beat-aware dilated cross-attention strategy. We further blend refined beat and music features to generate comprehensive multi-condition features. This strategy not only enhances beat and motion controllability in dance generation but also maintains the realism and coherence of the produced dance motions. Additionally, leveraging our beat representation, we introduce a novel beat alignment loss, explicitly ensuring synchronization between dance motions and the given beat condition, thereby significantly enhancing the overall quality of generation. Extensive experiments confirm that our method outperforms current state-of-the-art approaches in terms of beat alignment and motion controllability. Beyond the advantages above, our framework also supports arbitrary beat designation and flexible frame assignments of key poses.

In summary, our main contributions are fourfold:

- We introduce a multi-condition dance generation framework that achieves beat synchronization and enhanced motion controllability. To the best of our knowledge, our method makes the first attempt at beat-controllable key pose-guided dance generation.
- We present a hierarchical multi-condition fusion mechanism to effectively suppress the conflicts and fully exploit the complementary information among different conditions.
- We delve into the property of beats and formulate it as a nearest beat distance representation. A beat alignment loss is further tailored to offer explicit supervision signals to the generated dance motions, largely elevating the synchronization between the generated motions and the given beat conditions.
- Extensive experiments demonstrate our method performs favorably against the state-of-the-art approaches, especially in motion controllability and motion-beat alignment.

2 Related Work

2.1 Human Motion Generation

Human motion generation aims to generate realistic human motion automatically. Previous works can be generally categorized into three genres according to the input condition: text-conditioned [5, 18, 24, 56, 61], audio-conditioned, and scene-conditioned [4, 10, 16, 45]. Our work belongs to audio-conditioned human motion generation. But different from the primary audio-conditioned human motion generation, music-to-dance generation requires a comprehensive consideration of aesthetic movements, emotional expression, and precise synchronization with musical beats, making it extremely challenging to render convincing results. Recently, researchers have widely explored the application of diffusion models

in human motion generation [2, 3, 5, 16, 18, 61, 62], which shows unprecedented performance than traditional generative models. In this paper, we get a deep insight into the beat synchronization problem and investigate a multi-condition diffusion-based scheme for controllable dance generation.

2.2 Dance Generation

With the emergence of deep learning [47–49], dance generation has attracted lots of research interest in recent years. Previous methods have explored various frameworks with different types of backbones to achieve single-condition dance generation, including CNN [14], RNN [1, 15, 39, 51], GCN [7, 32], VAE [9, 38], GAN [19, 20, 37], and Transformer [21–23, 35, 36, 43, 53]. FACT [23] proposes a full-attention cross-modal transformer model to capture the correlations between music and dance. Danceformer [21] presents a two-stage deterministic framework for audio-driven single-condition dance generation. It first generates key poses from the input music and then performs interpolation between the key poses. However, it falls short of arbitrary beat choreography and flexible key pose controllability due to its single-condition nature. Bailando [35] utilizes separate VQ-VAEs on upper/lower half bodies and a motion GPT to map the music and seed poses to dance sequences. Several recent attempts have investigated the diffusion models for dance generation [29, 42, 55], which significantly pushes the boundaries of music-to-dance performance. Compared to applying only a single input condition, multi-condition dance generation incorporates additional conditions beyond music into the generation process for better controllability, such as dance style labels [2], text [11], and keyframes [42, 52]. For example, LDA [2] introduced dance style labels as auxiliary conditions. However, it is limited to generating pre-defined styles and lacks flexibility. TM2D [11] adopts a VQ-VAE based model to support dual-modal music and text-driven dance motion generation. Yang et al. [52] employ normalizing flows to model dance motion probability distributions and utilize time embedding to achieve keyframe control. EDGE [42] adopts a diffusion-based framework and enables key pose control via temporal/spatial blending during inference. Although the existing methods have investigated additional conditions for better dance generation control, they typically neglect explicit guidance and supervision to align the generated motion dances with specified beats. Consequently, these methods struggle to produce both beat-specific and key pose-guided dance sequences. In contrast, we propose to explicitly disentangle the beat condition from music and inject flexible beat and key pose controllability into the entire generation process, significantly enhancing the beat alignment while supporting flexible frame assignments of key poses.

2.3 Multi Condition Diffusion Generation

Multi-condition diffusion generative models seek to inject multi-condition guidance into the generation process for versatile controllability [6, 8, 25, 28, 31, 34, 44, 50, 54, 59, 60]. Benefiting from large-scale training data, Stable Diffusion [33]

constructs a powerful foundation for image generation. Building upon this, ControlNet [57] and T2I-Adapter [27] efficiently integrate trainable parameters into existing models, enabling the inclusion of spatially localized input conditions in a pre-trained text-to-image diffusion model. Uni-ControlNet [58] proposes a framework equipped with two additional control modules for multi-condition control. Qin et al. [30] present a unified model capable of handling various visual conditions, resulting in a more streamlined scheme. Codi [40] acquire multi-modal synthesis skills through the mapping of diverse modalities into a unified space, training on lists of different multi-modal generation tasks. The aforementioned works typically controllable generation using pre-trained models under diverse conditions. Our work, conversely, focuses on the effective and comprehensive fusion of multiple conditions by presenting a hierarchical multi-condition fusion mechanism aimed at better integrating sparse keyframe control conditions with dense music and beat control conditions, resulting in more controllable dance generation.

3 Method

3.1 Problem Definition

Dance generation aims to create realistic dance motion sequences $\mathcal{X} = \{\mathbf{x}^1, \mathbf{x}^2, \mathbf{x}^3, \dots, \mathbf{x}^L\}$ from a given piece of music $\mathcal{C}$ with a duration of L . Here, $\mathbf{x}^i \in \mathbb{R}^D$ represents a human pose at the i -th frame, which is denoted as a D -dimensional vector. In this paper, we endeavor to disentangle beat condition from the input music and simultaneously incorporate explicit beat and key pose guidance to achieve beat-synchronized, key pose-guided dance generation. Given a music condition $\mathcal{C}$, a sparse set of keyframes $\mathcal{X}^{ref} = \mathcal{X} \odot \mathbf{M}$, where $\mathbf{M} \in \{0, 1\}^L$ is a temporal binary mask that specifies the assigned locations of the keyframes, and a beat condition $\mathcal{B}$, our goal is to generate a beat-specific, and key pose-guided dance motion sequence $\hat{\mathcal{X}} = \{\hat{\mathbf{x}}^1, \hat{\mathbf{x}}^2, \hat{\mathbf{x}}^3, \dots, \hat{\mathbf{x}}^L\}$ that conforms to the music $\mathcal{C}$, consistent with reference frames $\mathcal{X}^{ref}$ at positions specified by $\mathbf{M}$, and synchronized with the beat condition $\mathcal{B}$.

3.2 Preliminaries

In this work, we adopt diffusion model [13, 28] as the backbone of our framework. Diffusion models establish a consistent Markovian forward process that incrementally introduces noise into clean sample data $\mathbf{x}_0^{1:L} \in q(\mathbf{x}_0)$, and a corresponding reverse process that progressively eliminates noise from noisy samples. For brevity, we use $\mathbf{x}_t$ to represent the entire sequence. During the forward process, a pre-defined noise variance schedule β_t is employed to regulate the noise increments at each step. The forward process can be formulated as follows:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}). \quad (1)$$

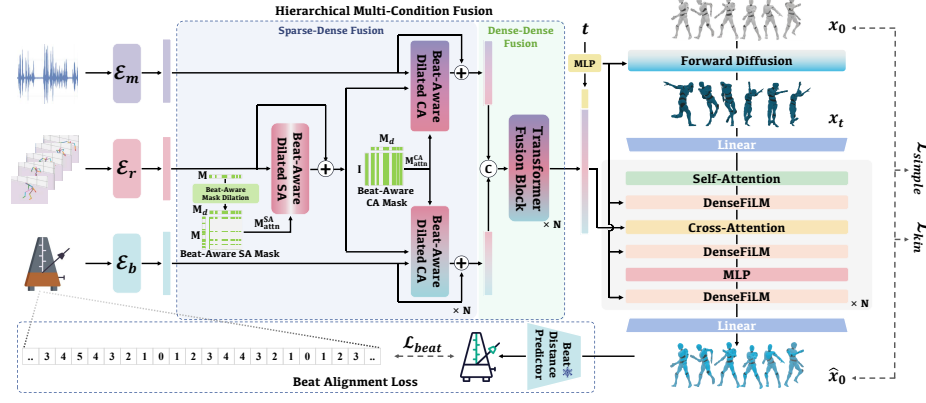


Fig. 2: Overview of our proposed method, *Beat-It*. We generate a beat-synchronized dance sequence utilizing music, keyframes, and beat conditions. Conditional embeddings are derived and subsequently fused in a two-stage process: initially integrating sparse keyframe condition with other dense conditions, followed by the fusion of these dense conditions. The final fused condition is then processed by the conditional diffusion module. To ensure precise beat control, a beat alignment loss is employed to explicitly supervise the generated motions at the beat level.

After T steps, the sample data progressively transforms into a noise distribution $q(\mathbf{x}_T)$, which is usually a standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In the reverse process, the noise is gradually removed from the noisy sample $\mathbf{x}_T$ to produce the clean sample $\mathbf{x}_0$. In our task, multi-conditional dance generation aims to model the distribution $p(\mathbf{x}_0|\mathbf{C})$ with a set of conditions $\mathbf{C}$. Following Ho et al. [13], we directly predict the clean sample $\mathbf{x}_0$ from the noise distribution $q(\mathbf{x}_T)$ with the following objective:

$$\mathcal{L}_{simple} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}|\mathbf{C}), t \sim [1, T]} [\|\mathbf{x}_0 - G(\mathbf{x}_t, t, \mathbf{C})\|_2^2]. \quad (2)$$

A spatial or temporal constraint $\mathbf{x}^c$ with positions specified by a binary mask m can be directly incorporated into the denoising process without additional training. This can be achieved by simply performing the following operation at every timestep:

$$\mathbf{x}_t = \mathbf{x}_t^c \odot m + \mathbf{x}_t \odot (1 - m), \quad (3)$$

where $\odot$ denotes the Hadamard product. By doing so, regions under constraint can be replaced with forward-diffused samples of the constraint, while the remaining regions are left unchanged. However, it is worth noting that this simplistic approach tends to render inferior results when given sparse constraints.

3.3 Overview

The pipeline of our proposed model is illustrated in Fig. 2. To begin with, we feed the music condition $\mathcal{C}$, keyframe condition $\mathcal{X}^{ref}$, and beat condition $\mathcal{B}$ into

three encoders respectively: the music encoder $\mathcal{E}_m$, keyframe encoder $\mathcal{E}_r$, and beat encoder $\mathcal{E}_b$. This process yields the corresponding music embedding $\mathbf{e}_m$, keyframe embedding $\mathbf{e}_r$, and beat embedding $\mathbf{e}_b$, where $\mathbf{e}_m, \mathbf{e}_r, \mathbf{e}_b \in \mathbb{R}^{L \times d}$. Then, we forward these condition embeddings to a hierarchical multi-condition fusion module to produce comprehensive multi-condition features $\mathbf{c}_f$, which are finally sent to the conditional diffusion denoising module to render the denoised dance motion sequence $\hat{\mathcal{X}}$.

3.4 Beat Representation

As mentioned in previous studies [23, 35, 42], there exists a strong consistency between musical beats and dance motion beats. In practical dance choreography, the arrangement of motion beats is relatively flexible and does not require a strict point-to-point alignment with musical beats. Therefore, we introduce an independent beat condition disentangled from the music, to allow for flexible motion beat controllability in dance generation. For an input music sequence of duration L , each frame can be categorized as either a beat or non-beat frame. A straightforward way to represent the beat condition is using a binary mask, where 1 and 0 denote the beat and non-beat frames respectively. However, such a simple binary presentation exhibits high sparsity as the beat frames only account for a small portion of the whole music sequence. This makes it less informative and, at times, perceived as noise or disregarded by the model. As an alternative, we propose to represent the beat condition as a vector b , with each entry $b^i \in \mathbb{N}$ denoting the distance between the current frame and the nearest beat frame. This representation not only mitigates the sparsity caused by binary formulations but also provides the model with local temporal context. This significantly facilitates the precise acquisition of rhythmic characteristics within the choreography and injects more effective beat controllability into the learning process. The beat embedding $\mathbf{e}_b \in \mathbb{R}^{L \times d}$ is obtained using a separate embedding layer, followed by the transformer-based encoder $\mathcal{E}_b$. Please refer to the supplementary materials for detailed representations of music and dance.

3.5 Hierarchical Multi-Condition Fusion

Although the keyframe constraint offers precise pose specifications for specific frames, it is still too sparse which significantly increases the learning difficulty of the model. Naively incorporating this condition with others introduces a vast of padding, leading to excessive noises that may impair the effectiveness of keyframe constraints. By contrast, music condition and beat condition are much more denser, with valid values for each frame. To harmoniously combine these three conditions, we introduce a novel multi-condition hierarchical fusion mechanism.

Specifically, in the first stage, we inject sparse keyframe condition into other dense conditions through a beat-aware dilation attention strategy, which shares a similar spirit of DiffKFC [46]. We observe that dance movements on motion-beat frames are often more iconic in the whole sequence, distinguished by their higher emphasis and significance. As these movements typically serve as guiding

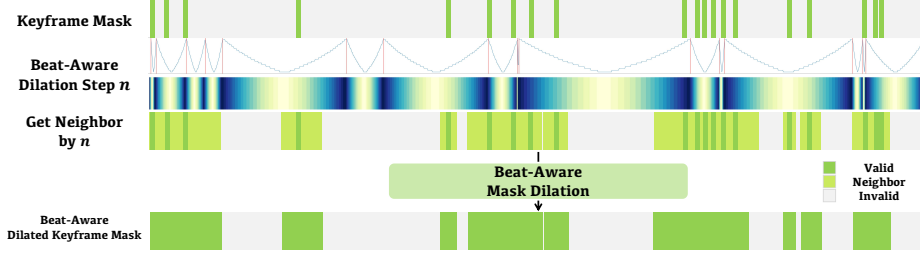


Fig. 3: Illustration of the beat-aware mask dilation scheme. The first row visualizes the keyframe mask, with deep green indicating valid control constraints and gray indicating invalid ones. The second row presents the dilation step curve with red lines marking beat frames. The third row is a heatmap of the dilation step. The fourth row shows the neighborhood range of keyframes, with light green indicating the expanded valid region from beat-aware mask dilation. The final row displays the beat-aware dilated keyframe mask.

elements, influencing the choreography and style of the entire dance, they exert a more substantial influence on the surrounding frames. To utilize this property, we design a beat-aware mask dilation scheme. Unlike DiffKFC which uses identical dilation steps for all keyframes, treating them uniformly, we opt for different dilation weights based on the distance between the keyframe and the nearest designated beat frame. The dilation step n is calculated as $n = \lceil s \cdot e^{-2 \frac{b^i}{d^i}} \rceil$, where b^i and d^i are the ground truth beat distance at frame i and the distance between frame i 's adjacent motion-beat frames, respectively. Base dilation step s is a hyper-parameter controlling overall dilation speed. The closer the keyframe is to the nearest beat frame, the larger the dilation step. This strategy enables the model to gradually propagate the keyframe condition to the surrounding frames with beat awareness. The beat-aware mask dilation scheme is illustrated in Fig. 3. Given the keyframe binary mask $\mathbf{M}$ and dilation step n , the beat-aware dilated mask $\mathbf{M}^d$ is obtained as follows:

$$\mathbf{M}_d[i] = \begin{cases} \max_j \mathbf{M}[i - j], j \in \{-n, n + 1, \dots, n - 1, n\} & \text{if } \mathbf{M}_i = 1, \\ \mathbf{M}_i, & \text{otherwise.} \end{cases} \quad (4)$$

The beat-aware dilated attention mask is then calculated as $\mathbf{M}_{attn} = \mathbf{M}\mathbf{M}_d^T$. This beat-aware mask dilation scheme enables beat-aware keyframe condition propagation, thereby amplifying the synergy among conditions characterized by varying degrees of sparsity, thus significantly boosting the generation quality.

In the second phase, we concatenate two refined dense conditions, namely keyframe-fused music embedding and keyframe-fused beat embedding. These concatenated conditions are fed into a transformer condition fusion module, yielding the final fused features that can be subsequently sent to the conditional diffusion denoising block.

3.6 Beat Alignment Loss

For controllable dance generation, it is crucial to ensure precise beat alignment between the generated dance motions and the given beat conditions. Current methods typically ignore the explicit constraints on the beats of the generated dance sequences. As a consequence, these approaches lack necessary beat controllability and struggle to guarantee beat synchronization between the generated dance motions and musical beats, which severely deteriorates the overall generation quality. To conquer this issue, building upon our beat representation, we present a beat alignment loss to provide explicit supervision that enforces the beats of the generated motions to precisely align with the given beat conditions. To this end, we first pre-train a beat distance estimator, which accepts a motion sequence as input and predicts the distance from the current frame to the nearest motion beat frame for each frame in the sequence. During the training process, the pre-trained beat distance estimator is responsible for providing supervision signals on the beats of the generated motions. The beat alignment loss $\mathcal{L}_{beat}$ can be formulated as follows:

$$\mathcal{L}_{beat} = \sum_{i=1}^L w_s^i \cdot w_b^i \cdot \mathbf{MSE}(b^i, \hat{b}^i), \quad (5)$$

where $w_s^i = \frac{1}{1 + e^{a \cdot (c - \|b^i - \hat{b}^i\|/b^i)}}$, and $w_b^i = e^{-\frac{2b^i}{d^i}}$. Here, $\hat{b}^i$ represents the predicted beat distance at frame i . w_s^i denotes an adaptive weight, which is derived from [26]. It aims to accentuate supervision in regions where beat alignment is imprecise. This is achieved by suppressing the loss magnitude in well-aligned beat regions. This strategy enables the model to focus more attention on challenging instances. The hyper-parameters a and c are used for controlling the degree of penalization and the threshold for beat alignment accuracy, respectively. $\mathbf{MSE}(\cdot, \cdot)$ denotes the mean squared error. w_b^i is an adaptive weight that enhances the supervision of frames closely aligned with the beats while diminishing the supervision of frames distantly related to the beats. Benefiting from the beat alignment loss, our model can yield dance motions with superior beat alignment and controllability, largely elevating the quality of the generated results.

3.7 Other Losses

Apart from the proposed beat alignment loss $\mathcal{L}_{beat}$, we also employ the basic diffusion loss $\mathcal{L}_{simple}$ (Eq. 2) and several additional auxiliary losses to govern the training.

Kinematic Loss. We adopt the same auxiliary losses as [41, 42]. The joint positions loss $\mathcal{L}_{joint}$ (Eq. 6) and velocities loss $\mathcal{L}_{vel}$ (Eq. 7) are used to improve the overall physical plausibility of the generated dance motions.

$$\mathcal{L}_{joint} = \frac{1}{L} \sum_{i=1}^L \|FK(\mathbf{x}_0^i) - FK(\hat{\mathbf{x}}_0^i)\|_2^2, \quad (6)$$

$$\mathcal{L}_{vel} = \frac{1}{L} \sum_{i=1}^L \|\mathbf{x}_0^{i'} - \hat{\mathbf{x}}_0^{i'}\|_2^2 + \|FK(\mathbf{x}_0^i)' - FK(\hat{\mathbf{x}}_0^i)'\|_2^2. \quad (7)$$

The contact consistency loss $\mathcal{L}_{contact}$ (Eq. 8) proposed in EDGE [42] is employed to alleviating the foot-sliding.

$$\mathcal{L}_{contact} = \frac{1}{L} \sum_{i=1}^L \|FK_{foot}(\hat{\mathbf{x}}_0^i)' \cdot \hat{g}^i\|_2^2, \quad (8)$$

where i denotes the index of the frame, and $FK(\cdot)$ is the forward kinematics function that converts the generated 6-DOF rotation representation into 3D key points in Cartesian space. $\hat{g}^i$ denotes the predicted binary foot contact label. In addition, we employ acceleration loss $\mathcal{L}_{acc}$ to further improve the quality of generated dance motions, which is suggested in previous work [35], preventing jitters in the generated dance motions.

$$\mathcal{L}_{acc} = \frac{1}{L} \sum_{i=1}^L \|\mathbf{x}_0^{i''} - \hat{\mathbf{x}}_0^{i''}\|_2^2 + \|FK(\mathbf{x}_0^i)'' - FK(\hat{\mathbf{x}}_0^i)''\|_2^2. \quad (9)$$

The overall kinematic loss is formulated as follows:

$$\mathcal{L}_{kin} = \lambda_{joint}\mathcal{L}_{joint} + \lambda_{vel}\mathcal{L}_{vel} + \lambda_{contact}\mathcal{L}_{contact} + \lambda_{acc}\mathcal{L}_{acc}, \quad (10)$$

where λ_{joint} , λ_{vel} , $\lambda_{contact}$, and λ_{acc} are hyper-parameters controlling the weights of the corresponding losses. We set $\lambda_{joint} = 1$, $\lambda_{vel} = 2.5$, $\lambda_{contact} = 10$, and $\lambda_{acc} = 0.1$ empirically. The overall loss function is formulated as follows:

$$\mathcal{L} = \mathcal{L}_{simple} + \lambda_{kin}\mathcal{L}_{kin} + \lambda_{beat}\mathcal{L}_{beat}, \quad (11)$$

where λ_{kin} and λ_{beat} are hyper-parameters controlling the weights of the corresponding losses. λ_{kin} and λ_{beat} are set as 1 and 0.5, respectively.

4 Experiments

Dataset. Following previous works [35, 42], we choose the most widely used dataset, AIST++ [23] for both training and testing. This dataset comprises 1,408 dance sequences encompassing 10 distinct street dance genres. Each dance genre incorporates 6 musical pieces with different BPMs. 85% of these sequences are categorized as basic choreographies, where performers execute the same choreography across all 6 musical pieces, each featuring varying tempos to synchronize with distinct musical beats. With its high choreographic and beat diversity, this dataset is ideal for our controllable dance generation task.

Implementation Details. For the hierarchical multi-condition fusion module, we implement six sparse-dense fusion blocks and two dense-dense fusion blocks, utilizing a base dilation step s set to 4, 8, 12, 16, 20, 24. For diffusion

process, we adopt the cosine schedule as beat schedule and set diffusion steps t as 1000 diffusion following [41]. The classifier-free guidance scale is set to 2. The diffusion module aligns with a transformer-based architecture similar to [42]. We adopt Adam optimizer with the learning rate set as $2e-4$ and the training batch size is set to 64. During training, we randomly sample between 1% and 30% of the ground truth (GT) motion frames as the keyframe conditions. This training strategy can effectively improve the generalization capability of our method on different numbers of keyframes. We use the GT motion beats as beat conditions.

To ensure fair comparisons, we follow EDGE [42] to crop all data into 5-second clips at 30 frames per second, with 2.5 seconds of overlapping. During testing, we randomly sample 10% of keyframes from unpaired samples in the testing set as keyframe conditions and subsets of testing musical beats as beat conditions to prevent data leakage. The musical beats used for experiments are extracted by the off-the-shelf audio toolkit Librosa [17]. All experiments are conducted on 4 NVIDIA RTX3090 GPUs.

Evaluation Metrics. For quantitative evaluation, we measure the generated dance from three aspects: generation quality, diversity, and controllability. Due to the limited samples in the testing set of AIST++ [23], the prior work EDGE [42] has demonstrated that the Fréchet Inception Distance (FID) is not reliable for measuring the generation quality. Therefore, we do not adopt FID as the evaluation metric. Instead, we assess generation quality from two different perspectives: music-dance correlation and kinematic plausibility. In terms of music-dance correlation, we utilize the Beat Alignment Score (BAS) metric, following the approach presented in [35]. BAS quantifies the synchronization quality between the generated dance and the musical beat. For the evaluation of kinematic plausibility, we adopt the Physical Foot Contact score (PFC) metric proposed in [42]. PFC provides a metric for measuring the physical plausibility of the generated dance from a kinematic perspective. As for the diversity, we measure the average feature distance of kinetic (Div_k) and geometric (Div_g) features extracted by fairmotion [12] following the previous works [23, 35, 42].

For assessing controllability, we employ the Key Pose Distance (KPD) and Beat Assignment Precision (BAP) metrics to evaluate keyframe control and beat control, respectively. KPD quantifies the average mean square error of local joint positions in Cartesian space at the keyframes. To evaluate beat control, we measure the alignment between the generated dance and the designated beat condition. BAP is defined as the percentage of beats in the generated dance that are correctly assigned to the designated beat frames. Note that this metric differs from the BAS, as a dance sequence with a high BAS does not necessarily adhere to the specific motion beat choreography constraint we intend to impose.

4.1 Comparison to Existing Methods

Currently, only three music-to-dance methods are publicly accessible: EDGE [42], Bailando [35], and FACT [23]. Of these, EDGE [42] is most directly aligned with our approach, as it similarly accommodates keyframes as inputs for generating dance sequences. We therefore assess EDGE using the same keyframe conditions

Table 1: Quantitative comparisons among different methods on AIST++. **Bold** indicates best result. $\downarrow$ means lower is better, $\uparrow$ means higher is better and $\rightarrow$ means closer to the ground truth is better.

Methods	Quality		Diversity		Controllability	
	PFC $\downarrow$	BAS $\uparrow$	$\text{Div}_k \rightarrow$	$\text{Div}_m \rightarrow$	KPD $\downarrow$	BAP $\uparrow$
Ground Truth	1.338	0.384	9.773	7.212	-	-
FACT [23]	2.698	0.202	9.704	7.342	-	-
Bailando [35]	1.578	0.215	9.622	7.175	-	-
EDGE [42](keyframes)	1.084	0.235	9.743	7.274	0.859	-
Ours(beat & keyframes)	0.966	0.661	9.660	7.248	0.306	0.793

as our method to ensure a fair comparison. Conversely, certain approaches, like DanceFormer [21], are not open-source, thus hindering any direct evaluative comparison. We use the original test split of AIST++ [23] for evaluation, which has no overlap with the training split in terms of both the music and dance choreography.

Table 2: Results of user study.

User Study	Beat-It Win Rate	
FACT	92.2%	
Bailando	78.8%	
EDGE (keyframes)	Quality	Controllability (Keyframe)
	60.3%	86.9%

Table 3: Ablation study on AIST++.

Ablations	Quality		Controllability	
	PFC $\downarrow$	BAS $\uparrow$	KPD $\downarrow$	BAP $\uparrow$
w/o HF	25.626	0.322	0.477	0.323
w/o BD	1.632	0.358	0.389	0.371
w/o $\mathcal{L}_{beat}$	1.342	0.397	0.343	0.411
Ours	0.966	0.661	0.306	0.793

Quantitative Comparisons. The quantitative results are shown in Tab. 1. As is revealed, our method surpasses all the competitors in terms of generation quality, diversity, and controllability. Particularly, our method shows considerable advantages over the compared methods for BAS metrics, notably with an improvement of 0.426 compared with the SOTA method EDGE [42]. This demonstrates the effectiveness of our method in generating beat-synchronized dance sequences while maintaining diversity. Furthermore, in the aspect of controllability, our method also significantly excels EDGE [42] with a notable improvement of 0.553 in KPD. Fig. 4 shows the mean joint velocities over time for a specific dance motion concerning both the GT and the existing methods. Notably, compared with others, our approach demonstrates more motion beats (local minimum point of velocities) precisely at the beat frames. This contributes to a better visual alignment with the audio when compared to the other methods. The BAP further validates our method’s superior ability to produce dance sequences precisely aligned with the beat condition. Due to limited space, we present additional quantitative experiments in the supplementary materials.

User Study. We also conduct a user study to assess the generation quality and controllability of our method. We randomly select 20 generated dance sequences

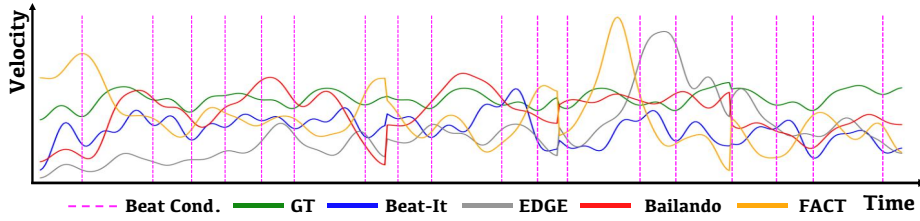


Fig. 4: Visualization comparison on beat alignment among different methods. The motion generated by our method shows precise beat alignment with the given beat condition, demonstrating the superiority of our method in beat control.

from the testing set and ask 18 participants to pick the best one among our method and the other compared methods. In particular, participants are required to choose the superior dance according to several criteria, namely, the overall visual plausibility of movements, synchronization with the dance beats, and keyframe controllability. The results of the user study are tabulated in Tab. 2. Our method shows noticeable advantages in terms of generation quality compared to other approaches, particularly in beat alignment. For qualitative evaluations on keyframe-conditioned controllability, we also provide the participants with the keyframe references for intuitive comparison between our method against EDGE [42]. The results show that our approach owns better motion controllability than EDGE [42].

4.2 Ablation Studies

To get a more comprehensive insight into our main contributions, we conduct ablation studies on the hierarchical multi-condition fusion module, the beat-aware dilation scheme, and the beat alignment loss to evaluate their efficacy. The quantitative results are shown in Tab. 3. The qualitative results can be also found in the supplementary video.

Hierarchical Multi-condition Fusion Module. To verify the significance of the hierarchical multi-condition fusion module, we remove it by simply concatenating all the conditions and feeding them into a one-stage encoder (w/o HF). As shown in Tab. 3, the simple one-stage fusion module can lead to inferior results in both quality and controllability. This is because the direct concatenation of multiple conditions can lead to inevitable condition conflicts, giving rise to suboptimal optimization of the model. Even worse, the naive padding of the sparse keyframe condition introduces severe noises, further complicating the learning process and resulting in unsatisfactory generation performance. By decently fusing multiple conditions gradually, our hierarchical multi-condition fusion mechanism effectively mitigates the conflicts among different conditions and yields more informative conditions, allowing for higher-quality generation with better controllability.

Beat-aware Dilation Scheme. To validate the effectiveness of our beat-aware dilation scheme, we replace it with a vanilla masked attention strategy (w/o

BD). As shown in Tab. 3, the absence of the beat-aware dilation scheme notably impairs both keyframe and beat choreography controllability. The reason behind this is that beat-aware dilation can dynamically adjust the expansion degree of the attention mask, contingent on the distance between the keyframe and the beat frames. This tactical approach significantly improves the synchronization between keyframes and their associated beat frames, thereby empowering the model to more effectively harness prior knowledge of beats. Consequently, this enhancement considerably bolsters beat control while augmenting keyframe controllability.

Beat Alignment Loss. To further validate the effectiveness of the beat alignment loss, we assess the model’s performance by comparing it to its counterpart without it (w/o $\mathcal{L}_{beat}$). As shown in Tab. 3, the ablated variant without the beat alignment loss demonstrated performance degradation in beat choreography, showcasing decreases of 0.264 and 0.382 in BAS and BAP, respectively. This decline in performance can be attributed to the advantageous impact of direct supervision during training, particularly in guiding the model to comprehend beat-level dance choreography through the proposed beat alignment loss. Consequently, this enhancement significantly improves the synchronization between motion sequences and designated beats.

5 Discussion and Conclusion

In this work, we propose a novel multi-condition diffusion-based framework, Beat-It, for beat-synchronized and key pose-guided dance generation. The proposed framework presents a hierarchical multi-condition fusion mechanism equipped with a beat-aware mask dilation scheme to integrate conditions with different information sparsity. To achieve precise beat synchronization, we explicitly disentangle the beats from the music and inject beat controllability throughout the entire generation process. Additionally, we introduce a specifically designed beat alignment loss to provide explicit guidance and supervision on motion beats. Both qualitative and quantitative experimental results on the AIST++ dataset validate the superiority of our method in producing high-quality dance sequences with precise beat synchronization and flexible keyframe control. Further limitations and broader impact are discussed in supplementary materials.

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